

GAIRA V7

Architecture Update After Phase 05

Supplementary architecture document · documentation only · no code, no algorithm, no artefact was changed

Date	2026-08-06
Branch	gaira-v7-rebuild
Phases complete	00 · 01 · 02 · 02.5 · 03 · 04 · 04.5 · 05
Phases archived on evidence	03 themes · 04 BSV · 04.5 Meta Components · 05 evidence axes
Canonical representation	49-dimensional CSM activation vector
Next phase	06 — Chemistry Evidence Layer (16-d)
V5 atlas	09ed804a40836f4a05a91ba10900cded — unmodified, in production
LSM registry	208482d6f7178b5b8f16cace91be55b0
CSM dictionary	0b4aa550ccefed3edabdbde5bae11c8d
Phase 05 engine	20d8bd99ce71f45a125c6a2b1d719e51
Corpus	375 spectra · 154 canonical molecules · 16 fine chemistry classes
Evaluation	v7_harness_v1 · 5 folds grouped by canonical_id · criteria FROZEN in Phase 00

THE ONE-SENTENCE SUMMARY

Six phases of evidence show that GAIRA's abstraction stack pays up to the Consensus Spectral Motif layer and not after it; three independent attempts to build a layer above it all lost information, so the CSM activation vector becomes the canonical representation and the interpretable layer is rebuilt with a label space that matches the project's own frozen evaluation ontology.

Sources of record

context/GAIRA_V7_ARCHITECTURE_STATUS_AFTER_PHASE05.md · architecture/GAIRA_V7_TARGET_ARCHITECTURE.md
plan/GAIRA_V7_REBUILD_PLAN.md · plan/SUCCESS_CRITERIA.md · plan/VALIDATION_AND_DECISION_RULES.md · context/TERMINOLOGY_AND

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including one that has stood since Phase 00
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learning mode · inference mode · validation · legacy/current/future · provenance

HOW TO READ THIS DOCUMENT

Nothing here was recomputed. Every number is quoted from a committed phase table and the table is named. If a number in this document disagrees with a phase table, the phase table is correct and this document is wrong.

Archived work is preserved, not deleted: every retired layer keeps its report, audit, figures, tables, artefacts and tests exactly as committed, and remains reproducible.

1 · The current architecture

GAIRA V7 today is a frozen projection engine over a 49-dimensional consensus motif basis. A spectrum is preprocessed on the canonical grid, projected by non-negative least squares onto the frozen CSM dictionary, and everything downstream reads that one vector. Nothing is fitted at inference; no output depends on which other spectra are in the batch.

layer	dim	what it does	status
canonical preprocessing	676	asLS → SG → L2, 450–1800 cm ⁻¹	frozen, unchanged from V5
LSM dictionary	50	class-local motifs, 16 independent fits	frozen
CSM dictionary	49	consensus motifs, 1 merge of 4 proposed	frozen · CANONICAL
reference bank	154×49	one activation vector per molecule	frozen
calibrator	—	Dirichlet, selected on Brier	frozen
rejection channels	8	residual, sparsity, entropy, distance ...	frozen
evidence axes	11	declared from band assignments	ARCHIVED, pending DG-06

What the engine delivers, measured on the frozen folds:

capability	measure	value	source
chemistry class, unseen molecule	top-1 / top-3	0.845 / 0.971	Phase 05
chemistry class, unseen molecule	macro F1	0.807	Phase 05
molecule identity, molecule present	top-1 / top-5	0.605 / 0.795	Phase 05
rejection of degraded spectra	joint AUROC	0.921	Phase 05
robustness, 7 × 5 perturbations	class retention	0.935 (raw 0.895)	Phase 05
provenance	chains broken	0 of 3,133	Phase 05
calibration	ECE / discrimination	0.130 / 0.891	Phase 05
reconstruction	mean explained variance	0.821	Phase 05

THE ONE NUMBER THAT DEFINES THE ARCHITECTURE

Chemistry-class top-1 on molecules the atlas has never seen rises 0.608 (raw spectrum) → 0.850 (LSM) → 0.855 (CSM), then falls to 0.664 (11 declared axes), 0.405 (themes / BSV) and 0.392 (Meta Components). The abstraction stack pays up to the CSM layer and not after it. Everything in §2 follows from that shape.

One honest caveat carried from Phase 05: molecule identification is a shortlist, not an answer. Top-1 of 0.605 on a 154-way problem is far above chance and is not identification. The engine should be read as a chemistry-class instrument that also returns a ranked molecular shortlist.

2 · What changed

The change is not cosmetic and it is not a change of taste. Four objects were built, measured against pre-registered criteria, and retired. Each remains on disk, fingerprinted and reproducible.

id	retired	measured	verdict
A-13	soft biochemical themes (4)	class top-1 0.855 → 0.405	archived
A-14	BSV = S^Tc (4)	inherits A-13; eff. rank 2.40 of 4	archived
A-15	Meta Components (3)	0.185 of CSM information retained	discarded
A-16	11 declared evidence axes	class top-1 0.664 vs CSM 0.845	archived*
A-17	SERS out-of-domain gate	AUROC 0.548 on real Ag-SERS	out of scope

* A-16 is archived pending DG-06. If the Chemistry Evidence layer does not clearly exceed it, the 11-axis profile is reinstated.

Added:

id	added	input	status
A-19	Chemistry Evidence (16-d)	CSM activation	Phase 06 · DG-06
A-20	BSV2 — biochemical programmes	Chemistry Evidence ONLY	Phase 07 · DG-07
A-21	hierarchical molecular retrieval	CSM + chemistry prior	Phase 08 · DG-08

Promoted:

A-08 — the 49-dimensional CSM activation vector becomes the canonical representation. It was always the projection basis; it is now also the object every downstream branch reads.

Amended principles:

- P-06 — chemical class remains an organisational prior and must never supervise a local fit. It is now admissible as an intermediate probabilistic evidence coordinate carrying uncertainty. It remains inadmissible as a terminal hard label.
- P-07 — unchanged in substance; 'theme' becomes legacy vocabulary and the principle now governs Chemistry Evidence axes and BSV2 programmes.
- P-18 (NEW) — stability without informativeness is not evidence. No layer, mode, calibrator or model may be selected on a reproducibility, stability or calibration metric without first clearing a pre-registered informativeness floor.

WHAT DID NOT CHANGE

Non-negativity at every layer. Learning offline, inference by projection. Determinism and fingerprint verification. Provenance as a first-class field. One canonical molecule = one reference unit. Class partitions the decomposition and never supervises it. The frozen V5 atlas remains the control arm, and the frozen Tier-1 success criteria are untouched.

3 · Why it changed

Phase 04 measured six abstraction levels on identical frozen splits. The question it answered was not 'is the representation good' but 'where does the hierarchy stop paying'.

level	dim	molecule top-1 (Split A)	class top-1, molecule unseen
L1 raw spectrum	676	0.790	0.608
L2 LSM	50	0.806	0.850
L3 CSM	49	0.799	0.855
L4 theme	4	0.553	0.405
L5 BSV	4	0.553	0.405
L6 geometry	5	0.495	0.541

Three separate constructions then occupied the slot above L3, by three different mechanisms:

- discovered by archetypal analysis over CSM co-activation — Phase 03 themes, 0.405;
- discovered by non-negative factorisation of the CSM activation matrix — Phase 04.5 Meta Components, 0.392, retaining 0.185 of the CSM layer's information;
- declared from Raman band assignments with no fitting at all — Phase 05 evidence axes, 0.664.

Declaring the axes rather than discovering them closed roughly half the gap. It did not close it. That is the strongest statement V7 has made about abstraction over this corpus: the loss is not an artefact of the discovery step.

THE PATTERN THAT MADE THIS HARD TO SEE — AND WHY P-18 EXISTS

Two of the three losing layers scored BETTER than the CSM layer on stability. The Phase 03 softmax theme mode was the most reproducible option available. Meta Components won every stability axis outright — replicate consistency 0.980 vs 0.893, activation-stability AURC 0.975 vs 0.928, class-retrieval AURC 0.944 vs 0.936 — while retaining 0.185 of the information.

A representation that says nearly the same thing about every spectrum is perfectly reproducible and useless. A fourth instance appeared in Phase 05 calibration, where the ECE-optimal calibrator reported 0.605 — the base rate — for every spectrum, with sharpness exactly 0.000.

Each was caught only after an explicit informativeness constraint was added. P-18 makes the constraint a precondition rather than a rescue.

The SERS retirement has a different cause and is worth separating. Phase 04's out-of-domain gate failed at AUROC 0.548 because a non-negative Raman-motif dictionary reconstructs Ag-SERS of the same metabolites comfortably. That is a true finding about the representation, not a defect — and it was answering a question the project does not ask. Raman-only scope removes it.

4 • Evidence for every change

Demonstrated — measured, and in most cases replicated across phases.

#	finding	value	source
D-01	corpus, folds, metrics, V5 control reproducible	byte-identical atlas	Phase 00
D-04	local motifs rarely describe the same phenomenon	1 merge of 1,225 pairs	Phase 02
D-05	motif space is a continuum with one bipartition	modularity 0.620 vs null 0.070	Phase 02.5
D-06	the abstraction stack pays up to the CSM layer	0.608 → 0.850 → 0.855	Phase 04
D-08	class inference generalises to unseen molecules	0.845 top-1, macro F1 0.807	Phase 05
D-09	CSM is more accurate AND more robust than raw	0.845 / 0.935 vs 0.592 / 0.895	Phase 05
D-11	provenance complete end to end	0 broken of 3,133 chains	Phase 05
D-13	degraded and structureless spectra are rejected	joint AUROC 0.921	Phase 05

Falsified — tested against pre-registered criteria and did not hold.

#	claim	measurement	verdict
F-01	themes are the semantic axes of the BSV	0.855 → 0.405	falsified
F-02	the BSV is the canonical output coordinate	eff. rank 2.40 of 4	falsified
F-03	second-order NMF recovers a programme layer	0.185 information retained	falsified
F-04	declaring the axes avoids the information loss	0.664 vs 0.845	partly falsified
F-05	the atlas detects real Ag-SERS as out-of-domain	AUROC 0.548	falsified
F-06	ECE $\leq$ 0.10 is achievable for the confidence	0.130 informative / 0.080 constant	falsified

Unknown — not measured by any phase so far.

#	unknown	addressed by
U-01	does a 16-d chemistry layer preserve what 4-, 11- and 3-d layers did not?	Phase 06
U-02	how much of the class signal is an imprint of the class partition? (R-01)	Phase 06 control
U-03	does the engine reject genuinely novel chemistry?	Phase 06 holdout
U-04	does BSV2 escape the pattern that discarded three predecessors?	Phase 07
U-05	does a chemistry prior beat direct cosine on molecule identity?	Phase 08
U-06	does V7 clear the frozen Tier-1 bar? NEVER MEASURED	Phase 06 / 09
U-08	does the CSM layer earn its place over the LSM layer? 0.845 vs 0.848	Phase 06

THE MOST UNCOMFORTABLE ROW IN THIS TABLE

U-02. The Chemistry Evidence layer predicts the same sixteen classes that partitioned the Phase 01 local decompositions. Risk R-01 has been flagged since Phase 00 and has never been controlled. Until the class-agnostic decomposition control is run, 0.845 cannot be described as a property of the representation rather than of the partition. DG-06 makes the control mandatory.

Figure A1 · GAIRA V7 learning mode

Where every object is learned, and what is frozen. Offline only — this path never runs at inference.

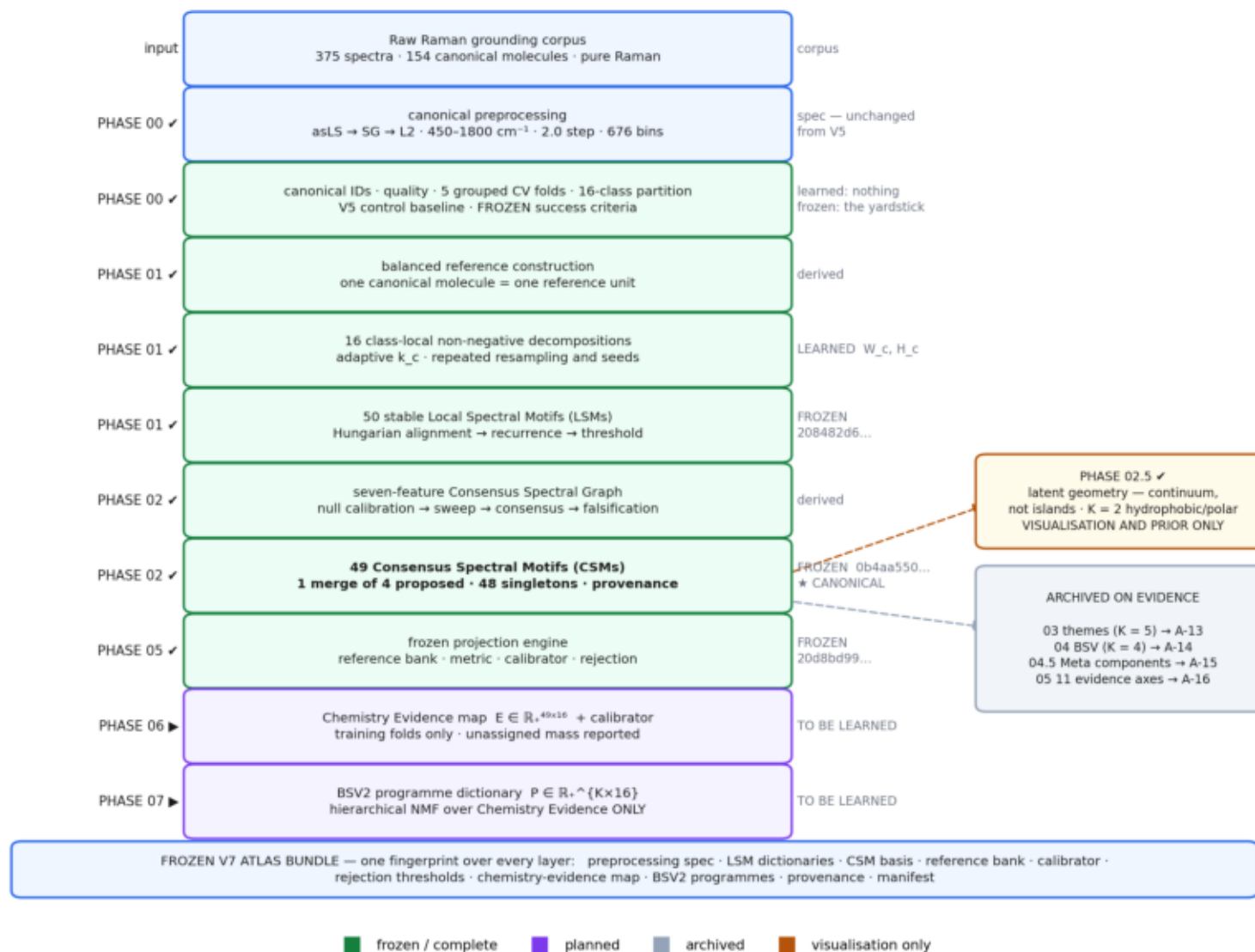


Figure A1 · Learning mode. Every fitted object and the phase that produced it. Green is complete and frozen; purple is planned; grey is archived on evidence; amber is visualisation only. Fingerprints are the committed values checked on every load.

Figure A2 · GAIRA V7 inference mode

Projection and arithmetic only. Nothing is fitted; no output depends on which other spectra are in the batch.

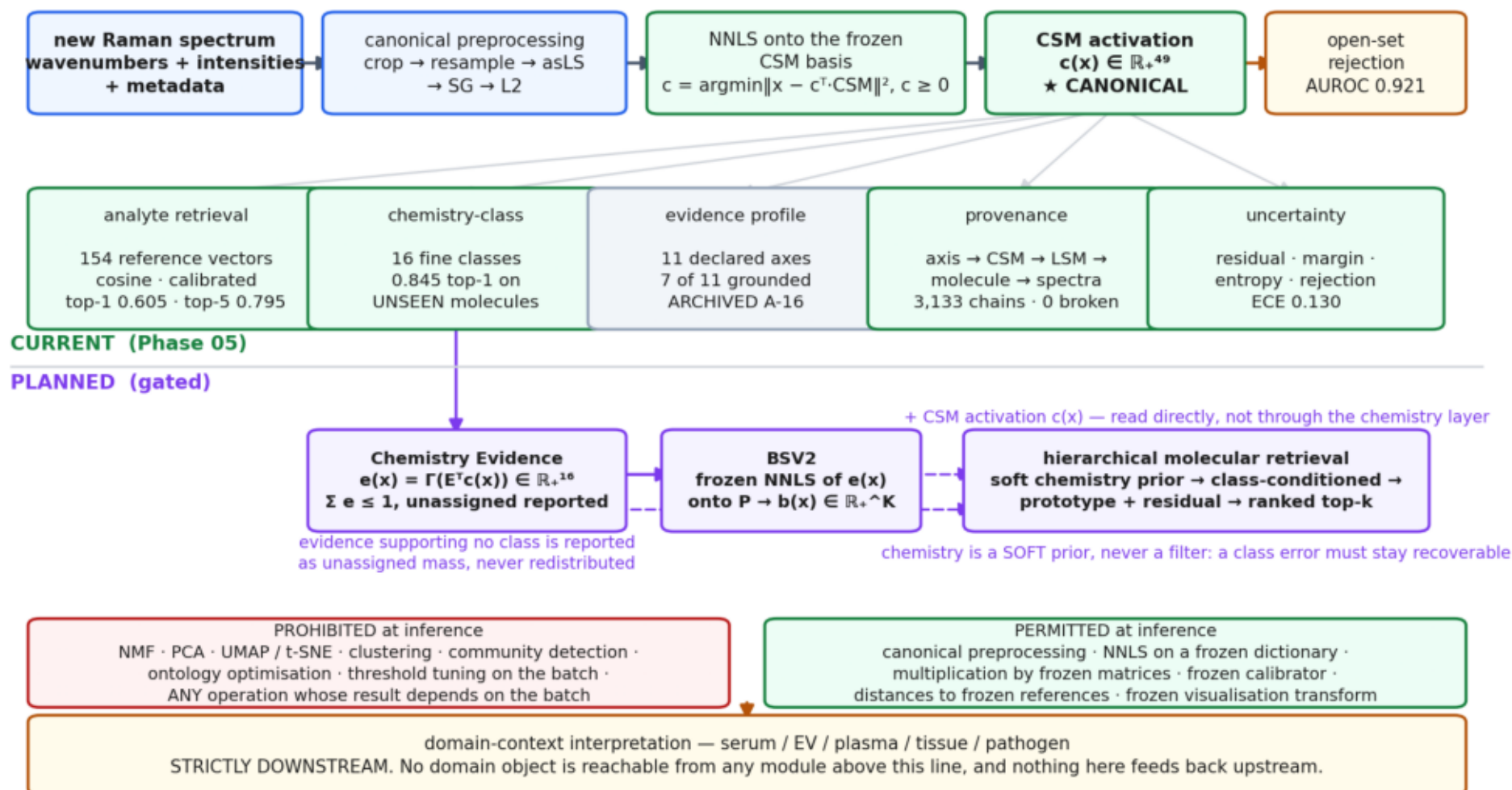


Figure A2 · Inference mode. Above the rule is what runs today; below it is what is planned, and every planned stage is gated. Note that molecular retrieval reads the CSM activation directly — the chemistry layer sits beside the representation, not instead of it. That is the structural lesson of the archived path.

Figure A3 · GAIRA V7 validation pipeline

Every remaining phase ends with the same four-part gate. A gate can send the phase back or change the architecture.

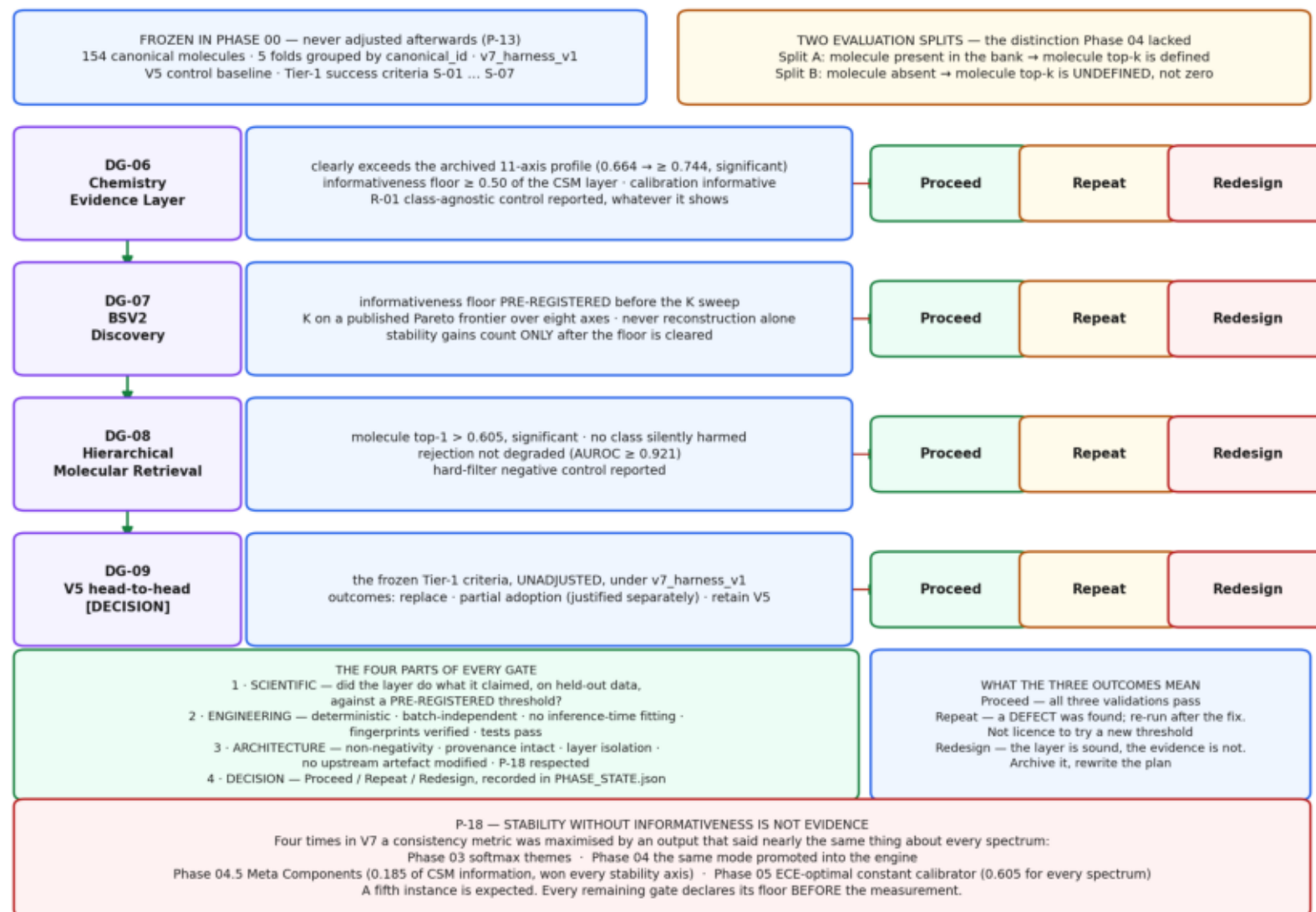
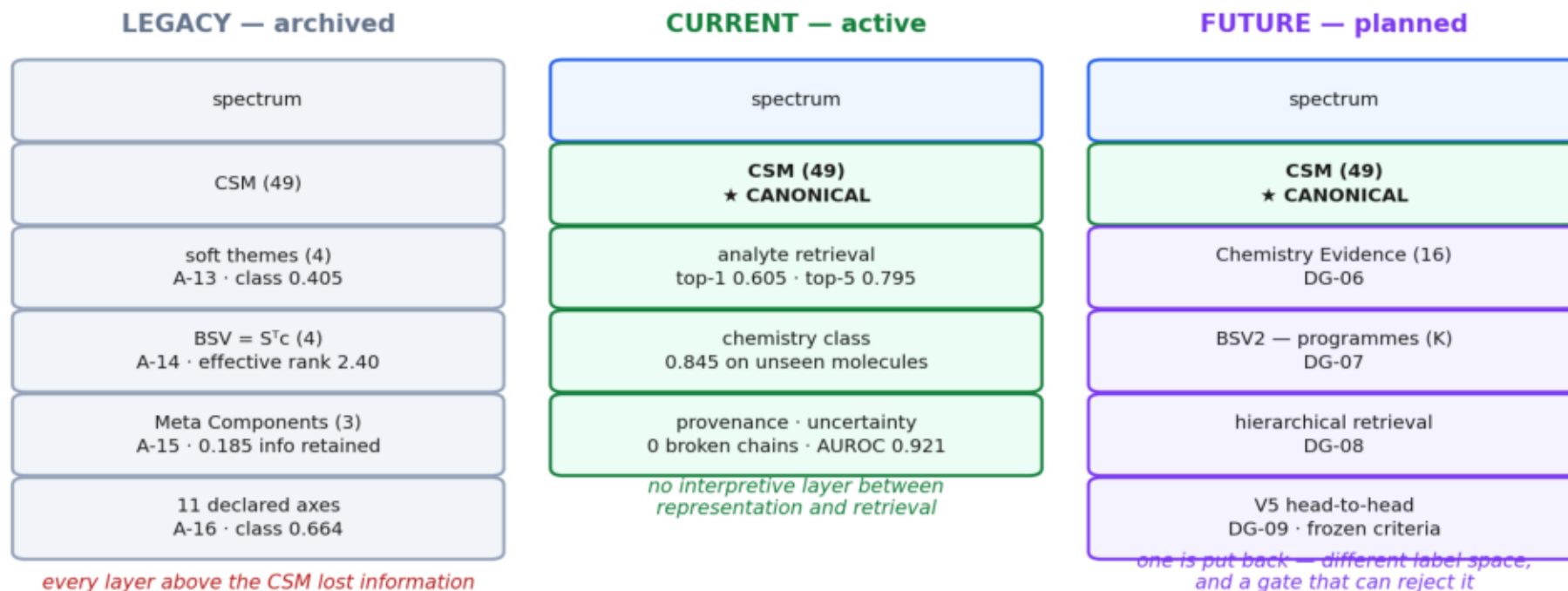


Figure A3 · Validation pipeline. Four remaining gates, each with the same four parts and the same three possible decisions. The P-18 banner records the four occasions on which a consistency metric was maximised by an uninformative output.

Figure A4 · Legacy, current and future architecture

The legacy path is preserved, not deleted. Every retirement is an evidenced decision, not a change of taste.



every layer above the CSM lost information

one is put back — different label space, and a gate that can reject it

THE EVIDENCE THAT RETIRED THE LEGACY PATH — chemistry-class top-1 on molecules the atlas has never seen, identical frozen folds

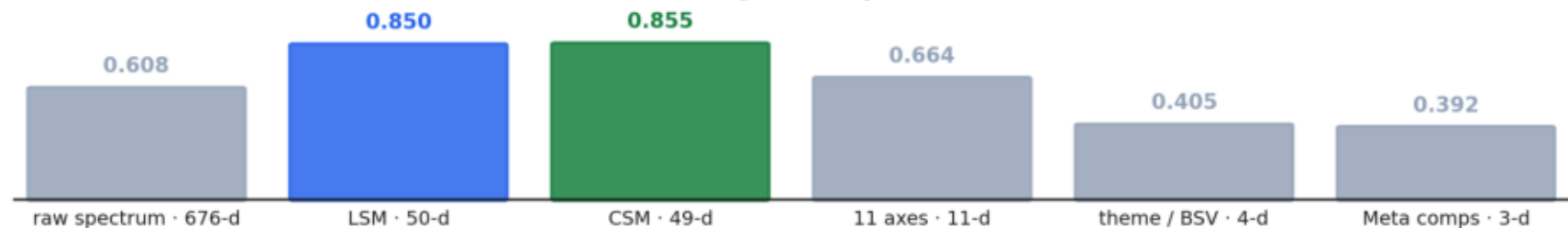
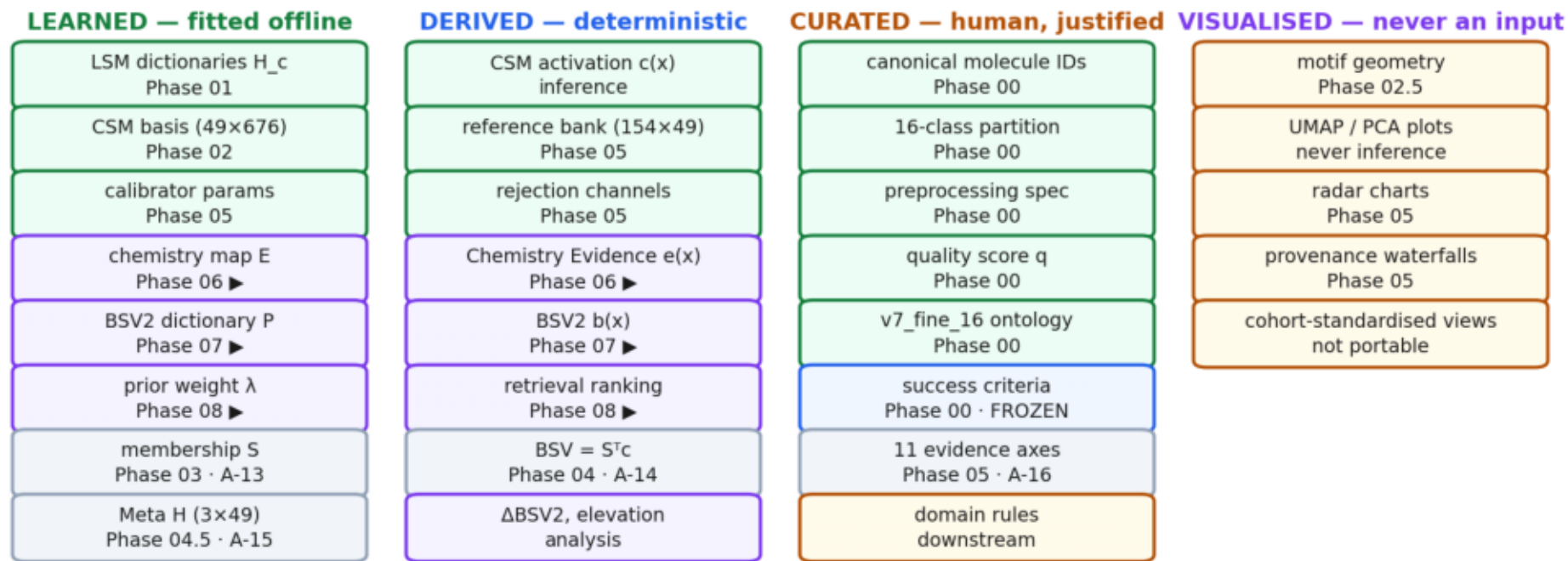


Figure A4 · Legacy, current and future architecture side by side, with the evidence that retired the legacy path. Chemistry-class top-1 on molecules the atlas has never seen, identical frozen folds: the curve rises to the CSM layer and falls after it.

Figure A5 · Where every object comes from

Learned offline · derived deterministically · curated by a human · visualised only. Nothing crosses from right to left.



THE ONE-WAY RULE

Curated objects may organise a fit but never supervise it (P-06, amended). Learned objects are frozen before inference (P-09). Visualised objects are never an input to interpretation, retrieval or scoring. Domain context is applied last and never feeds back upstream.

Figure A5 · Object provenance. Learned, derived, curated and visualised objects, with the one-way rule that governs how they may interact.

6 • The remaining roadmap

phase	objective	gate	the bar
06 Chemistry Evidence	frozen calibrated map $\mathbb{R}_+^{49} \rightarrow \mathbb{R}_+^{16}$	DG-06	> 0.744 vs 0.664
07 BSV2 Discovery	hierarchical NMF over evidence only	DG-07	≥ 0.50 info floor
08 Hierarchical Retrieval	soft chemistry prior on molecule ID	DG-08	> 0.605 top-1
09 V5 head-to-head	the replacement decision	DG-09	frozen Tier-1
10 Chemistry-aware learning	deferred	—	—
11 Corpus expansion	deferred	—	—

PHASE 06 — Chemistry Evidence Layer

Learn a frozen map $E \in \mathbb{R}_+^{49 \times 16}$ and a calibrator, on training folds only. Sixteen is not a hyperparameter: it is the size of the evaluation ontology frozen in Phase 00, which is also the label space of the frozen success criteria.

Deliverables: top-1/3, macro F1, balanced accuracy, per-class precision and recall, confusion matrix, calibration (ECE, Brier, sharpness, discrimination), replicate consistency, cross-fold stability, noise robustness, radar examples, provenance chains, the R-01 control, and the first measurement against the frozen Tier-1 bar.

DG-06 proceeds only if the layer CLEARLY exceeds the archived 11-axis profile.

PHASE 07 — BSV2 Discovery

Hierarchical NMF over Chemistry Evidence ONLY — never CSM activations. That restriction is what makes BSV2 a different object from the discarded Meta Components: it factorises chemistry co-occurrence rather than motif usage.

$K \in \{2,3,4,5,6,8,10,12,14\}$, selected on a Pareto frontier over eight axes: reconstruction, held-out chemistry prediction, programme stability, interpretability, mutual information, noise robustness, calibration, compression. Never reconstruction alone; never accuracy alone; the full frontier is published.

Output: biochemical programmes — not themes, not manual mappings.

PHASE 08 — Hierarchical Molecular Retrieval

Inputs: CSM activation AND Chemistry Evidence as a soft prior. Implement the soft prior, class-conditioned retrieval, prototype + residual scoring, hierarchical ranking, top-k with confidence, and conformal sets only if exchangeability is defensible under molecule-grouped splits.

The hard constraint: a class error must be recoverable. At 0.845 class accuracy, a hard filter would remove the correct molecule for roughly one spectrum in six before scoring began. Any design that cannot recover from a class error is rejected at the gate regardless of its mean accuracy.

PHASE 09 — V5 head-to-head [RETAINED]

The Tier-1 criteria were frozen in Phase 00 and HAVE NEVER BEEN MEASURED. Phase 05's 0.845 is a per-spectrum number on 5-fold grouped CV; the frozen bar is a per-analyte number at $n = 167$ under v7_harness_v1. Dropping the phase that measures the frozen bar would make the bar unreachable and nullify P-13 in practice.

7 • Scientific risks

RISK 1 — BSV2 IS THE FOURTH ATTEMPT AT THE SAME ARCHITECTURAL POSITION

Discovered themes, discovered meta-components and declared evidence axes all lost class information, and two of them scored better on stability while doing so. BSV2 differs in its input, and that is a real difference — but the position is identical and so is the failure mode to watch for.

Mitigation: the informativeness floor is pre-registered before the K sweep, K is chosen on a published Pareto frontier, and 'BSV2 does not improve on Chemistry Evidence' is an accepted publishable outcome.

RISK 2 — CIRCULARITY IN THE CHEMISTRY EVIDENCE LAYER (R-01, U-02)

The layer predicts the same sixteen classes that partitioned the Phase 01 fits. If the partition imprinted itself on the representation, part of 0.845 is an artefact of the experimental design.

Mitigation: DG-06 makes the class-agnostic decomposition control mandatory, and the gap must be published whatever it shows.

RISK 3 — 'OPEN-SET REJECTION' DOES NOT YET MEAN NOVELTY (U-03)

Every Phase 05 negative is corruption or structureless signal, four of six drawn from the same perturbation module used in the robustness study. The engine is shown to reject spectra corrupted by processes it was separately shown to be robust against. That is internally consistent and it is not evidence about novel chemistry.

Mitigation: a held-out-chemistry-class experiment in Phase 06. Until it exists, the phrase used must be 'rejection of degraded and structureless spectra'.

RISK 4 — THE CSM LAYER MAY NOT EARN ITS PLACE OVER THE LSM LAYER (U-08)

Class top-1 0.845 vs 0.848; retention 0.935 vs 0.923. Phase 02 accepted one merge of 1,225 candidate pairs, so 48 of 49 CSMs are single LSMs and the two layers are nearly the same object. Nothing measured so far justifies the CSM layer on performance; the justification is provenance and interpretability, and that should be stated rather than implied.

RISK 5 — THE CORPUS, NOT THE ARCHITECTURE, MAY BE THE BINDING CONSTRAINT (R-17)

375 spectra and 154 molecules is a small corpus for a 154-way retrieval problem and for calibration with ten confidence bins. If the remaining phases deliver small gains, the honest conclusion may be that further architectural work is not the lever, and Phase 11 corpus expansion is the correct next step. The plan is explicitly able to reach that conclusion.

A note on band shape (U-07). Amide I and cis C=C both sit near 1650 cm^{-1} and window-based reasoning cannot separate them; this is the leading cause of the amide axis's failure in Phase 05. Band width, asymmetry and splitting carry assignment information the architecture currently discards. It is the highest-value addition not on the roadmap.

8 • Expected outcomes

Written in advance so that the outcome cannot be reinterpreted afterwards (P-12). These are expectations, not predictions to be defended.

DG-06 — Likely to pass, with a caveat.

The 16-d layer has a larger label space than the 11 declared axes and matches the evaluation ontology exactly, so exceeding 0.664 is plausible. The caveat is U-02: if the R-01 control shows a large gap, the gain is partly the class partition talking to itself, and the report must say so. Passing DG-06 with an uncontrolled R-01 would be worse than failing it.

DG-07 — More likely to fail than to pass.

Three predecessors in the same position all fell below the floor. BSV2's different input is a genuine difference, but 16 → K is a compression of an already-compressed object. If it fails, Chemistry Evidence becomes the terminal interpretable layer, Phase 08 proceeds without BSV2, and V7 will have four independent measurements saying the same thing about abstraction over this corpus. That is a stronger scientific result than a marginal pass.

DG-08 — Genuinely uncertain, and the most interesting gate.

Molecule top-1 of 0.605 with class accuracy of 0.845 is exactly the configuration in which a soft prior should help: the chemistry is known far better than the identity. The risk is not that the prior fails but that it helps on average while harming specific classes — which is why S-34 requires every harmed class to be named.

DG-09 — Cannot be predicted, and should not be.

The frozen bar is +8 points on fine-16 retrieval over V5's 0.6707. Phase 05's 0.845 is suggestive but is a different protocol on a different unit of analysis. The honest position is that the measurement has not been made. If V7 clears it, V5 is replaced; if not, V5 is retained and the negative result is published in full.

The V7 rebuild has so far produced more negative results than positive ones, and that is not a failure of the programme. Phase 02 rejected three of four proposed motif merges. Phase 03 rejected one of five themes. Phase 04.5 discarded its entire output. Phase 05 left a gate failing rather than relaxing it. The one large positive result — 0.845 chemistry-class top-1 on molecules the atlas has never seen, with 0.935 robustness retention and complete provenance — is credible in proportion to how many opportunities there were to report it dishonestly.

9 • Inconsistencies found and resolved

THE MATERIAL ONE — THE FROZEN CRITERIA MEASURE A TASK THE TERMINOLOGY FORBIDS

SUCCESS_CRITERIA.md, frozen in Phase 00: S-01 requires 'CSM/MSS-equivalent fine top-1 ≥ 0.7507 ' against a V5 baseline of 0.6707. The baseline table shows the level it comes from:

v5_atlas::v7_fine_16 — retrieval of the correct fine-16 CHEMISTRY CLASS.

TERMINOLOGY_AND_DEFINITIONS.md: chemical class 'is not the inference output. V7 never predicts class.'

Both have stood since Phase 00. They cannot both be right.

RESOLUTION — the frozen thresholds stay exactly as they are; P-13 forbids adjusting them and nothing here does. The terminology was wrong: it conflated class as a TERMINAL CLAIM (rightly forbidden) with class as SUPERVISION INSIDE A FIT (rightly forbidden) and class as an INTERMEDIATE PROBABILISTIC EVIDENCE COORDINATE, which was never the danger and is what the frozen bar has always measured. P-06 is amended accordingly.

This does not make the 16-d layer safe. It resolves the conflict. U-02 remains the open scientific question.

Smaller inconsistencies, corrected in the same pass:

found

'Nothing in V7 has been implemented' in two documents
corpus cited as 167 analytes throughout

original vs canonical phase numbering mixed across documents
dependency map showed Phase 05 = BSV, Phase 06 = integration
phases/ directory names use original numbering
Phase 04 GATE_FAILED left as an open item

correction

corrected — six phases complete
canonical figure is 154 molecules from 375 spectra;
167 was the pre-audit normalised-name count
canonical numbering used, mapping restated
rewritten
retained (committed links) with a mapping table
recorded as out of scope under Raman-only (A-09/A-17)

ONE ITEM LEFT OPEN FOR THE PROJECT OWNER'S DECISION

The brief that commissioned this update specified the remaining roadmap as Phases 06, 07 and 08 only. Phase 09 — the V5 head-to-head against the frozen Tier-1 criteria — has been RETAINED rather than dropped, because the frozen criteria have never been measured and dropping the phase that measures them would make the bar unreachable and nullify P-13 in practice. If the intent was to retire the V5 replacement decision entirely, that is a legitimate call, but it should be made explicitly and recorded, not effected by omission.

ONE POLICY CONFLICT, RESOLVED IN FAVOUR OF THE MORE RECENT INSTRUCTION

A standing instruction from Phase 02.5 onward is 'PNG only, no SVG'. This update explicitly requested publication-quality SVG for the architecture diagrams. Both formats are written: SVG as requested, PNG at 200 dpi so the PDF embeds the same image the figure script produced and the per-phase convention is unbroken.

Provenance

claim class	source
all V7 phase numbers	results/v7_rebuild/<phase>/tables/ and artifacts/
the six-level hierarchy	phase04/tables/hierarchy_retrieval_v1.csv
Meta Component comparison	phase04_5/artifacts/verdict_v1.json
Phase 05 retrieval, calibration, rejection	phase05/artifacts/phase05_summary_v1.json
robustness retention	phase05/tables/robustness_summary_v1.csv
evidence-axis grounding and sensitivity	phase05/tables/evidence_axis_*.csv
V5 baseline	phase00/tables/phase00_baseline_metrics.csv (v7_harness_v1)
principles and risks	context/SCIENTIFIC_DESIGN_PRINCIPLES.md, plan/RISK_REGISTER.md

Fingerprints

object	fingerprint
V5 atlas (control, unmodified)	09ed804a40836f4a05a91ba10900cded
LSM registry	208482d6f7178b5b8f16cace91be55b0
CSM dictionary	0b4aa550ccefed3edabdbde5bae11c8d
theme registry (archived)	f54d4835ffdf8aa2d50a4a203da0e8f4
Phase 05 inference engine	20d8bd99ce71f45a125c6a2b1d719e51

SCOPE OF THIS UPDATE

Documentation only. No algorithm was implemented, no code modified, no inference changed, nothing retrained. The changes are confined to architecture, context, planning, terminology, pipeline diagrams, phase definitions, decision gates and success criteria.

No frozen artefact was touched. No frozen success threshold was altered. Every archived phase retains its complete output.

If a number in this document disagrees with a phase table, the phase table is correct and this document is wrong.